

Malek Ahmad | majobs112002@gmail.com | +972549498961 | [Linkedin](#) | [Github](#) | [Portfolio](#)

Backend Software Engineer specializing in **Python** and **JavaScript**, with hands-on experience **designing and building scalable, maintainable server-side systems**. Strong proficiency in **Python**, **backend architecture**, **API development**, and **database design**, with a solid foundation in **performance-oriented** and **systems programming**. Highly **motivated and fast learner**, collaborative **team player** with **strong problem-solving skills**, driven to build reliable, efficient, and scalable backend solutions.

Skills

Programming languages: C, C++, Java, Python, JavaScript.

Operating Systems: Linux, Windows.

Frameworks & Runtime: Node.js, Express.js, NestJS, Flask.

Concepts: Server-side development, RESTful APIs, authentication & authorization, scalable backend architecture.

Database: SQL (MySQL), SQLite, Firestore, MongoDB.

IDEs: Eclipse, CLion, PyCharm, VS Code.

Education

2021 –2025: Azrieli College of engineering | Software engineering | GPA 90.83

Courses: Distributed Systems | Introduction to Software Engineering | Software Engineering in the Service of the Community | Data Structures and Algorithms | Operating Systems | Computer Networks | Databases

Projects

- **2025: Retrieval-Augmented Generation (RAG) System for Multilingual Document Search**, [Code link](#)

Built a scalable **RAG web application** for internal use in academic institutions, enabling accurate Q&A over large, interconnected document collections in **Hebrew, Arabic, and English**.

Developed the **backend using Python (Flask), LangChain, and FAISS**. Designed a **React-based frontend** with a chat interface and file upload system.

- **2025: Reinforcement Learning AI for SOS Game**, [Code link](#)

Developed a self-learning AI agent for the SOS game **using Monte Carlo Tree Search (MCTS) and Puct algorithms**.

Built **using Python, TensorFlow/Keras**.

- **2024: Trainers' app for the Jerusalem Chess Center**, [Code link](#)

Developed a **React Native app** as part of a **4-person team**, connecting to a **Firestore database** to manage teachers, student groups, schools, and lessons.

Implemented backend logic for **CRUD operations, data synchronization, and real-time updates**.

Collaborated on **feature planning, architecture, and testing** to ensure reliable and scalable functionality.

- **2024: Project Management Web App**, [Code link](#)

Created a **server–client system in JavaScript** with **full CRUD operations** for projects, team members, and images from the **Unsplash API**. The data is stored in a **JSON file** acting as the **local database** in the server.

Certificates

- **2025: Place-IL Developers Internship Program**

- Successfully passed a **competitive** selection process (**250+ candidates**), including **coding challenges** in **OOP, data structures, and databases**.

- Successfully completed **HR, technical, and group** interview stages.

- **2024: CS50: Introduction to Artificial Intelligence with Python – Harvard University (edX)** [Projects and Certificate](#).

- Developed practical projects such as **image classification, sentiment analysis, Sudoku solver, and reinforcement learning agents**.

- Gained hands-on experience in **AI concepts**, including search algorithms, knowledge representation, optimization, and machine learning.

Experience

2022 – 2025: Tutor | Azrieli College of Engineering | Jerusalem

Tutored **4-5 students per year** in **programming, algorithms, math** and **physics** courses, improving my responsibility, self-confidence, and socialization skills.

Languages

Arabic (Native), Hebrew (Fluent), English (High level).